WealthWise: What the Web App and API Do

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What WealthWise does in one page

WealthWise lets an advisor open a client's investment account, see whether it still matches the client's plan, preview a fix, and place trades, all from three web screens backed by one API.

Screen	Address	What you use it for
Portfolios	/	See every account with its value, gain and how far it has drifted from plan
Portfolio detail	/portfolios/{id}	Inspect one account, preview rebalancing, place a trade, see its recent trades
Activity	/activity	Read the stream of trade events as they were recorded

The web app is a window onto the API. Everything you see is fetched from it, and everything you do (propose, trade) is sent to it. The API holds the rules, keeps the data and sends each trade on to a record-keeping queue.

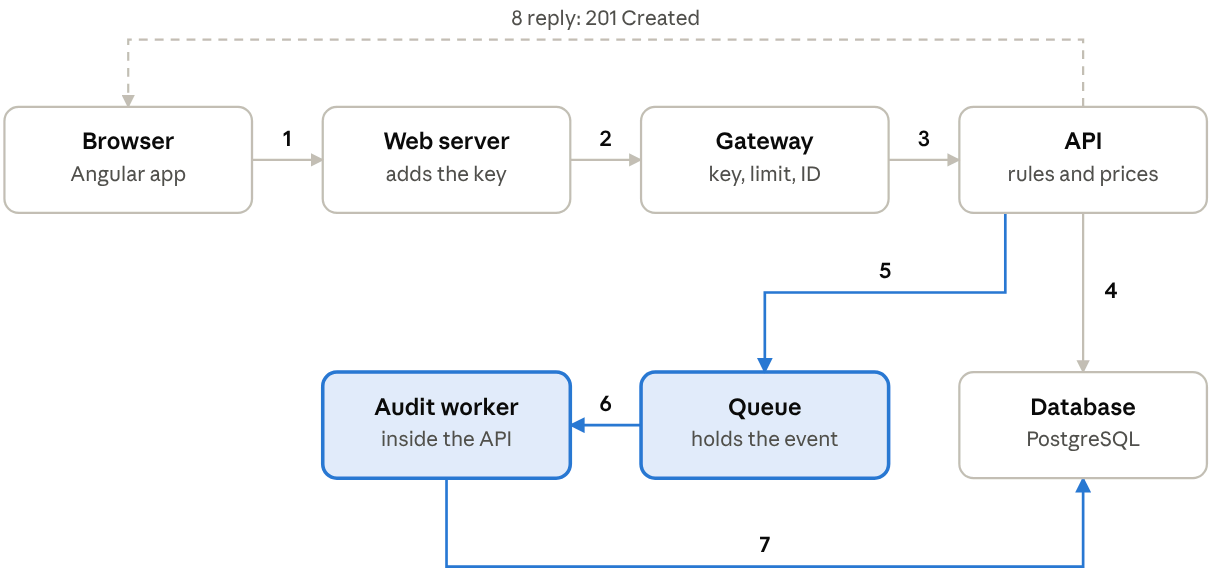
Where to open it. With Docker Compose the app is at http://localhost:4200. On the local Kubernetes cluster it is at http://localhost:8080. Both show the same three demo accounts, which start with different data because each has its own database.

The demo data. Three accounts (Harlow Family Trust, Okafor Retirement IRA, Lindqvist Growth Account) and seven generic funds. Prices are simulated and drift gently, so numbers move a little every few seconds. Nothing here is real market data or investment advice.

How one click travels

When an advisor places a trade, the order passes through five stations, is saved, and only then is announced to the record-keeping side of the system.

A trade is saved first, then announced: the audit entry follows the reply



Blue path: happens after the save, off the critical path. Dashed line: the reply to the browser.

the journey of one trade, 8 numbered steps

1. The browser sends the order.
2. The web server adds the gateway key and forwards it.
3. The gateway checks the key and the request limit, tags the request with an ID, and passes it on.
4. The API checks the rules, fetches the price and saves everything to the database in one step.
5. Once saved, the API puts a trade.executed note on the queue.
6. The audit worker, which runs inside the API, picks the note up.
7. It writes one audit row to the database, ignoring any repeat.
8. Meanwhile the reply (201 Created) has already gone back to the browser, so the advisor never waits for steps 5 to 7.

Reading screens works the same way without the blue path: the browser asks, the API calculates from the database and the price board, and the answer returns along the dashed route.

Web app tour: the Portfolios page

The Portfolios page is the advisor's morning view: one row per account, sorted by name, with a red flag on any account that has drifted more than 5 percentage points.

Around every page. A dark bar on the left carries the WealthWise name, links to Portfolios and Activity, and a reminder that the data is demo data with simulated prices. On a phone the bar becomes a strip across the top.

The table, column by column.

Column	What it shows	How to read it
Account	The account name (a link) and the client's name beneath it	Click the name to open that account
Risk profile	Conservative, Balanced or Growth	The style of mandate the client agreed to
Value	Market value of all holdings plus cash, in whole dollars	Updates when the page loads
Unrealized	Today's value of holdings minus what was paid for them	Green when positive, red when negative; nothing is sold, so the gain is on paper
Largest drift	The biggest gap between any asset class and its target, ignoring direction	Grey up to 5 points; red and bold above 5

The sentence under the heading counts the accounts and reminds you what drift means. While data loads the table says Loading; if the API cannot be reached, a red banner says so.

What you should see with the demo data (at the starting prices; live prices shift each figure slightly):

Account	Value	Largest drift	Where the drift is
Harlow Family Trust	about $600,000	about 13.2 points	Bonds are 13 points under target
Lindqvist Growth Account	about $400,000	about 7.3 points	Bonds are 7 points under target
Okafor Retirement IRA	about $250,000	about 27.4 points	Cash is 27 points over target

All three are flagged red, which is the point of the demo: each has something to fix.

Web app tour: the portfolio detail page

The detail page answers four questions about one account: what is it worth, how far has it drifted, what would fix it, and what has been traded. On a wide screen the left column shows the account and the right column holds the tools; on a phone everything stacks.

Header

- A link back to all portfolios, then the account name, and the client and mandate beneath it.
- The total value in large type, then three facts: **Cash**, **Cost basis** (what was paid for what is still held) and **Unrealized** gain or loss (green or red).

Allocation against target (the drift strip)

This is the centre of the page. There is one row for each asset class: Equities, Fixed income, Cash and Alternatives.

Part of a row	Meaning
Dark bar	The class's actual share of the account today
Blue vertical tick	The target share from the client's plan
Hollow ring (only after you propose trades)	Where the share would land if the proposal were carried out
Number on the right	Actual share, then "of" the target, for example 59.4% of 50%
Bar turns red	Drift is larger than the tolerance band (the band typed in the Rebalance box, or 5% when the Full strategy is chosen)

As well as colour, each bar carries a text description for screen readers, for example "Equities is 59.4 percent, target 50.0 percent".

Holdings

One row per security owned: symbol and fund name, asset class, quantity, current price, market value, and gain or loss against cost. Largest holding first.

Recent trades

The latest 15 trades, newest first: date and time, the order in words (for example Buy 10 VTI), the price paid and the total value.

Rebalance box (right column)

1. Choose a **Strategy**: *Threshold* (trade only classes outside the band) or *Full* (return everything to target).
2. For Threshold, set the **Band** in percentage points, from 1 to 25 (default 5).
3. Click **Propose trades**. Nothing is traded yet.
4. Read the result: lines such as "Sell Equities $56,085" and "Buy Fixed income $78,960", then **Cash afterwards**. The drift strip now shows rings for the projected result. If every class is within the band, the box says nothing needs doing.

Trade ticket (right column)

1. Pick a **Security** from the list of seven.
2. Choose **Buy** or **Sell**.
3. Enter a **Quantity** (must be above zero; fractions are allowed).
4. Click **Execute at market**. The price is whatever the market feed says at that moment.
5. A green message confirms the fill, for example "Buy 10 SGOV filled at 100.46". A red message explains a refusal. After a fill, the page reloads the account and its trade list and clears any old proposal.

Automatic refresh. While the page is open it re-prices the account every 15 seconds, so values and the drift strip move with the simulated market. The trade list refreshes after your own trades.

Web app tour: the Activity page and error messages

The Activity page is the audit trail: every executed trade appears here once, a moment after it happens, newest first, up to 40 entries.

Column	What it shows
Received	The time the trade event was created, to the second

Column	What it shows
Event	The event type, currently always trade.executed
Order	The trade in words, for example Buy 5 SGOV
Notional	The total money value of the trade

The page does not update by itself. Click **Refresh** after placing a trade. Until a trade has been made it says "No events yet". The entries are not read from the trade table: they arrive by a separate route (a message queue), which is why they lag by a second or two and why they prove that route works.

Messages you may see, and what they mean.

Message	What happened
The API is unreachable. Is the stack running?	The browser could not reach the API at all: the stack is stopped or starting
The gateway rejected the API key.	The gateway returned 401; the key it is given is missing or wrong
Rate limit reached at the gateway. Try again in a moment.	More than 120 requests in a minute; wait briefly
Insufficient cash: order needs X, available Y.	A buy costs more than the account's cash
Cannot sell N SYMBOL: position is M.	A sell is larger than the shares held
Quantity must be positive.	The quantity was zero or negative
Portfolio … was not found.	The account address does not exist
Request failed (number).	Anything else; the number is the HTTP status

A stale-data conflict (two changes to one account at the same instant) shows a technical message from the database layer, because the screen displays the API's detail text; reloading the page and trying again resolves it. Showing a friendlier message there is a small planned improvement.

API reference

The API offers eight calls under /api and a ninth address that describes the API itself, and everything it returns is plain JSON.

Conventions. Calls through the gateway need an `apikey` header (the web app adds it for you). Money is a number in dollars. Ratios are fractions, so 0.05 means 5%. Names such as Buy or Equity travel as words. Times are UTC. Errors come back in a standard shape with a `title`, a `status` and a readable `detail`.

Method and path	What it does	You send	You get back
GET /api/health	Says the service is alive	nothing	200 with a status and the time
GET /api/securities	Lists everything that can be traded	nothing	200: symbol, name, asset class, base price for each of the 7 securities
GET /api/portfolios	Lists accounts for the first screen	nothing	200: id, name, client, risk profile, total value, unrealized gain, largest drift
GET /api/portfolios/{id}	One account with live valuation	the account id	200: header details plus holdings and the allocation table; 404 if unknown
GET /api/portfolios/{id}/rebalance	Proposes trades without making them	optional `strategy` (threshold or full) and `band` (a fraction between 0 and 1)	200: the plan; 422 for an unknown strategy; 404 if unknown account
GET /api/portfolios/{id}/trades	Lists past trades, newest first	optional `take` (default 25, 1 to 200)	200: the trades; 404 if unknown account

Method and path	What it does	You send	You get back
POST /api/portfolios/{id}/trades	Places a trade at the current price	a JSON body with symbol, side, quantity	201 with the trade; 404, 409 or 422 on failure
GET /api/audit	Lists recorded trade events, newest first	optional `take` (default 25, 1 to 200)	200: event id, type, time, payload
GET /openapi/v1.json	The machine-readable description of all of this	nothing	200: OpenAPI document

A plain `/health` also exists on the API's own port for container health checks; it is not routed through the gateway.

Example: place a trade. Illustrative values; prices vary by the second.

```
POST /api/portfolios/{id}/trades
{ "symbol": "SGOV", "side": "Buy", "quantity": 5 }

201 Created
{ "id": "6f0c...", "symbol": "SGOV", "side": "Buy", "quantity": 5,
  "price": 100.45, "notional": 502.25, "executedAt": "2026-11-
04T19:20:11+00:00" }
```

Example: a refused trade.

```
422 Unprocessable
{ "title": "Business rule violated", "status": 422,
  "detail": "Insufficient cash: order needs 28,940,999,710.59, available
58,772.75." }
```

Example: propose a rebalance (Harlow Family Trust, threshold strategy, band 0.05, starting prices).

```
GET /api/portfolios/{id}/rebalance?strategy=threshold&band=0.05
{ "strategy": "threshold", "totalValue": 600000.00,
  "trades": [
    { "assetClass": "Equity",      "side": "Sell", "amount": 56085.00,
"reason": "Drift outside the ±5% band." },
    { "assetClass": "FixedIncome", "side": "Buy",  "amount": 78960.00,
"reason": "Drift outside the ±5% band." } ],
  "cashAfter": 36400.00,
  "projectedAllocation": [ { "assetClass": "Equity", "actual": 0.5, "target":
0.5, "drift": 0.0 }, "..." ] }
```

Status codes at a glance.

Code	Meaning here	Who says it
200, 201	Success, or a trade created	API
401	Missing or wrong key	Gateway
404	No such account	API
409	The account changed while you were acting; retry	API
422	A business rule refused the request	API
429	More than 120 requests in a minute	Gateway
500	A genuine fault; reported and logged as an error	API

How the numbers are calculated

Every figure on screen comes from five small calculations, all done in the API's Domain project and all checkable with a calculator.

Figure	How it is worked out
Market value of a holding	Quantity times current price, rounded to cents
Total value of an account	All market values plus the cash
Unrealized gain	Market value minus cost basis, holdings only (cash is not included)

Figure	How it is worked out
Actual share of a class	Value of the class divided by the total; cash, and any fund classed as Cash, count in the Cash class
Drift	Actual share minus target share; positive means too much, negative too little

Cost basis uses the average-cost method. Buying adds the money spent. Selling removes a proportional slice: selling 4 of 10 shares removes 40% of the cost. Selling everything clears it.

Largest drift on the Portfolios page is the biggest drift in size, ignoring whether it is too much or too little.

A worked example. Harlow Family Trust at the starting prices is worth $600,000, of which $59,275 is cash.

Asset class	Value	Actual	Target	Drift	Trade needed to reach target
Equities	$356,085	59.35%	50%	+9.35 points	Sell $56,085
Fixed income	$131,040	21.84%	35%	-13.16 points	Buy $78,960
Cash	$59,275	9.88%	5%	+4.88 points	Funds the trades above
Alternatives	$53,600	8.93%	10%	-1.07 points	Buy $6,400

The two rebalancing strategies on that account.

- **Full** moves every non-cash class to its target. It sells $56,085 of equities and buys $78,960 of fixed income and $6,400 of alternatives. Cash afterwards: $59,275 - $78,960 - $6,400 + $56,085 = **$30,000**, exactly the 5% target. Trades under $50 are ignored.
- **Threshold** with a 5-point band trades only classes that are *more than* 5 points off: equities and fixed income. Alternatives (1.07 off) are left alone. Cash afterwards is **$36,400**, or 6.1%, a little above target because the Alternatives purchase was skipped.

Good to know.

- Cash is never traded directly. It is the pocket the money comes from or goes to, which is why the box reports **Cash afterwards**.
- The comparison is "more than", so an account sitting at exactly 5.00 points off is not traded at a 5-point band. The Okafor account's Alternatives are in precisely that position.

- Prices move, so proposals are recalculated from the current prices every time you click Propose trades.

Walkthroughs

Three short stories show the system end to end: a trade that works, a trade that is refused, and how the audit trail stays honest.

1. An advisor buys 10 shares of SGOV for Harlow Family Trust

1. **On screen.** In the trade ticket the advisor picks SGOV, Buy, quantity 10, and clicks Execute at market.
2. **Over the wire.** The browser sends the order to the web server, which adds the gateway key and passes it to the gateway. The gateway checks the key, counts the request against the 120-per-minute limit and stamps it with a correlation ID. The API receives it.
3. **Price.** The API loads the account and asks the price board for SGOV. Suppose it says $100.45.
4. **Rules.** The cost is 10 times $100.45 = $1,004.50, and the account holds $59,275 in cash, so the order is allowed.
5. **Changes.** Cash falls to $58,270.50; a new holding of 10 shares costing $1,004.50 is created; a trade record is added; the account's version number goes up by one.
6. **Saved together.** All of that is written to the database in one step: either every change lands or none does.
7. **Reply.** The API answers 201 Created. The screen shows a green "Buy 10 SGOV filled at 100.45" message, reloads the account and the trade list, and clears any old proposal. Because SGOV is a cash-style fund, the account's *Cash share* barely moves.
8. **Afterwards, separately.** The API drops a trade.executed note on the queue. Within a few seconds the consumer writes it into the audit log, and it shows on the Activity page after a Refresh.

2. A trade that is refused

The advisor asks to buy 1,000,000 shares of VTI. At about $285 a share that costs roughly $285 million; the account has $59,275.

- The rule check fails before anything is changed.

- The API answers 422 with "Insufficient cash: order needs ..., available ...", and the screen shows it in red.
- **Nothing is saved, no event is sent and no audit row appears.** The API log gets a single quiet line, not a stack trace, because a refused trade is normal business, not a fault.

3. Why the audit trail can be trusted

What goes wrong	What the system does
The queue delivers the same note twice	Each note has a unique ID; the second copy is recognised and ignored, so five trades always give five rows
Several API copies read the queue at once	They share the work; the unique ID still guarantees one row per trade
The queue is briefly unavailable when a trade is made	The trade stands and a warning is logged. The audit entry for that one trade is missing: a known gap, planned fix is a transactional outbox
Two people trade on one account at the same instant	The second request is told the account changed (409) and can retry

Behind the scenes

Six quiet mechanisms make the screens behave, and none of them needs the advisor to think about it.

1. Simulated prices. There is no real market feed. Each security has a base price (for example $285.40 for VTI) and a gentle, repeating wave added to it, so prices drift up and down predictably over minutes. The swing is about 3% for shares, 3.5% for property and gold, 1.2% for bonds and 0.04% for the cash fund, so the cash fund stays near $100.45.

2. A five-second memory. The API remembers each price for 5 seconds. That way the figure on screen and the price a trade is executed at agree, instead of differing by a tick.

3. The front door.

Protection	What it does
Key check	Requests without the right `apikey` get 401
Rate limit	More than 120 requests in a minute get 429
Cross-origin rule	Only pages served from the app's own address may call the API from a browser
Correlation ID	Every request gets a unique number, returned in a header, so one request can be followed through the logs
Key kept on the server	The web server adds the key, so it is never in the code the browser downloads

4. The message queue. After a trade is saved, the API posts a small envelope on a queue: an ID, a type (trade.executed), a time and the trade details. A background worker inside the API waits up to 5 seconds at a time for messages, takes up to 10, records each in the audit log, and only then removes the message. If the worker dies mid-way, the message becomes visible again after 15 seconds and is retried, which is why the audit write must be safe to repeat. Locally the queue is ElasticMQ; on AWS the same code talks to Amazon SQS.

5. Safety built into the data. Besides the rules in code, the database itself enforces that every holding and trade refers to a real security and account, that an account holds each security only once, and that each audit event ID is unique. A version number on the account catches two simultaneous changes.

6. Starting up and staying up. On first start the API waits for the database (retrying for about a minute), builds the tables, and adds the three demo accounts and seven securities only if none exist. Health checks let the platform restart a stuck container, and routine refusals are logged quietly so real errors stand out.

Demo data, limits and what the app does not do

The demo ships with three accounts and seven funds so that every feature has something to show; each account starts deliberately off-plan.

The accounts.

Account	Mandate	Targets (shares / bonds / cash / alternatives)	Opening holdings	Cash left
Harlow Family Trust	Balanced	50 / 35 / 5 / 10	VTI 900, VXUS 1,500, BND 1,800, VNQ 300, GLD 100	$59,275
Okafor Retirement IRA	Conservative	30 / 55 / 10 / 5	VTI 250, BND 900, VTIP 400, SGOV 300	$63,435
Lindqvist Growth Account	Growth	75 / 10 / 3 / 12	VTI 700, VXUS 1,200, BND 150, GLD 250, VNQ 100	$35,265

Each account's opening purchases appear in its trade history as past trades, a week apart.

The securities (illustrative funds and base prices; the live price wobbles around these):

Symbol	Name	Class	Base price
VTI	US Total Stock Market ETF	Equities	$285.40
VXUS	International Stock ETF	Equities	$66.15
BND	US Aggregate Bond ETF	Fixed income	$72.80
VTIP	Short-Term Inflation-Protected ETF	Fixed income	$48.90
VNQ	Real Estate ETF	Alternatives	$91.30
GLD	Gold Trust	Alternatives	$262.10
SGOV	0-3 Month Treasury ETF	Cash	$100.45

What the app deliberately does not do.

- It moves no real money and talks to no broker.
- It has no individual log-in; one shared key protects the API.
- It does not execute a whole rebalance in one click; proposals are advice and trades are placed one at a time.
- It proposes amounts per asset class, not which specific fund to trade, and ignores tax and fees.

- It handles one currency (US dollars) and uses simulated prices.
- The Activity page shows the latest 40 events and does not refresh itself.

Starting over. Resetting the data (see the operations scripts in the project) wipes the database and recreates exactly the demo data above.